HHS3190M
HUMAN PHYSIOLOGY AND FUNCTIONAL ANATOMY FOR REHABILITATION SERVICES
HHS3190MJ
復康服務之人體生理學及功能性解剖學

Physiology L5 Digestive System

生理學 L5：消化系統

(Sem 1 AY26-27 26-27學年第1學期)

Outline大綱

- Overview of digestive system
  消化系統概述
- Functional anatomy of digestive system
  消化系統的功能性解剖學
- Physiology of chemical digestion and absorption
  化學性消化和吸收的生理學

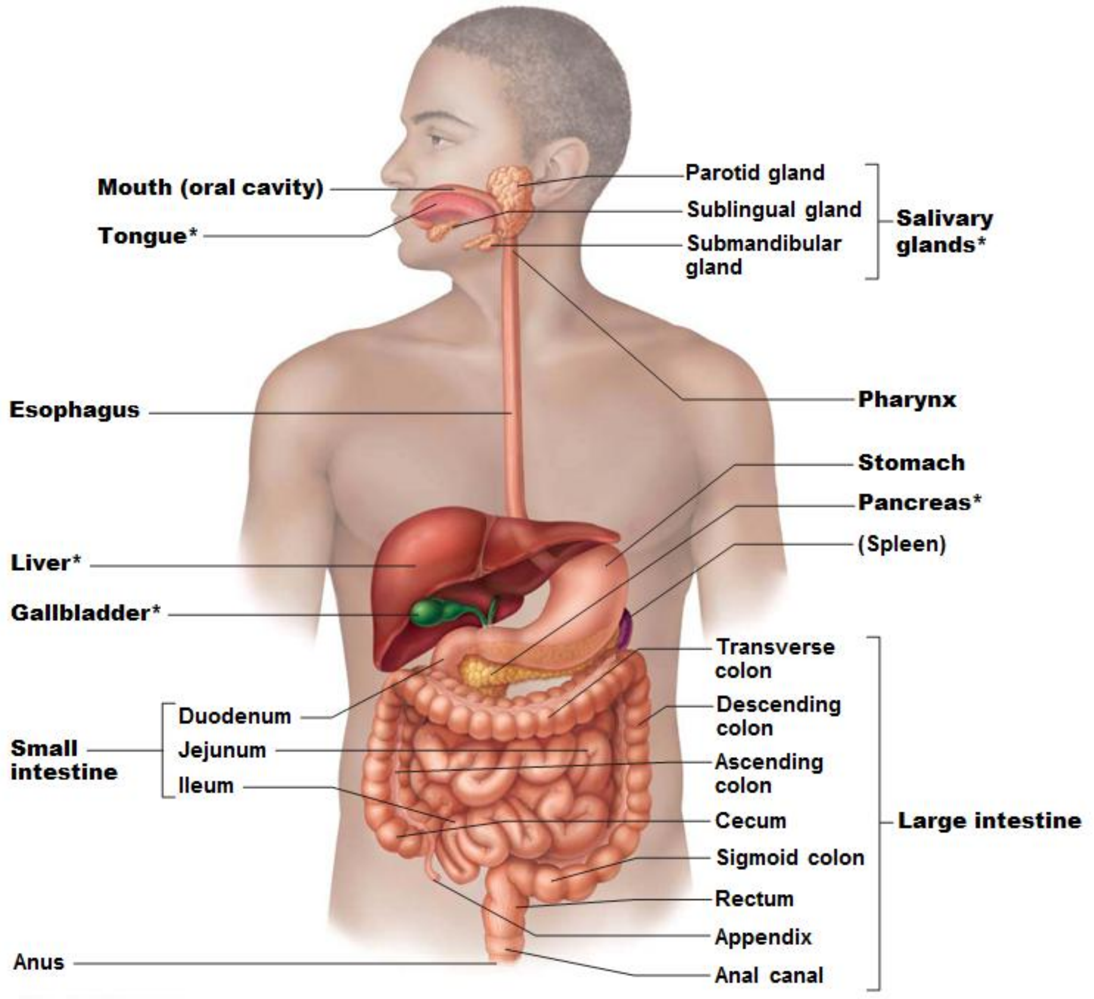
Overview of Digestive System
消化系統概述

- A healthy digestive system is essential to maintain life because it converts foods into raw materials that build and fuel our cells.
  健康的消化系統對於維持生命至關重要，因為它將食物轉化為構建和為細胞提供能量的原材料。

- Main functions of the digestive system 消化系統的主要功能
- **Ingestion 攝入**: Take in food 攝入食物
- **Digestion 消化**: Break it down into nutrient molecules 將其分解成營養分子
- **Absorption 吸收**:
  Absorb molecules into the bloodstream 將分子吸收到血液中
- **Egestion 排泄**:
  Rid body of any indigestible remains 清除體內難以消化的殘留物

Overview of Digestive System
消化系統概述

- Organs of the digestive system fall into two groups: 消化系統的器官分為兩類:
- (1) **Alimentary canal 消化道** (gastrointestinal or GI tract or gut 胃腸道或腸道)
- (2) **Accessory digestive organs 輔助消化器官**

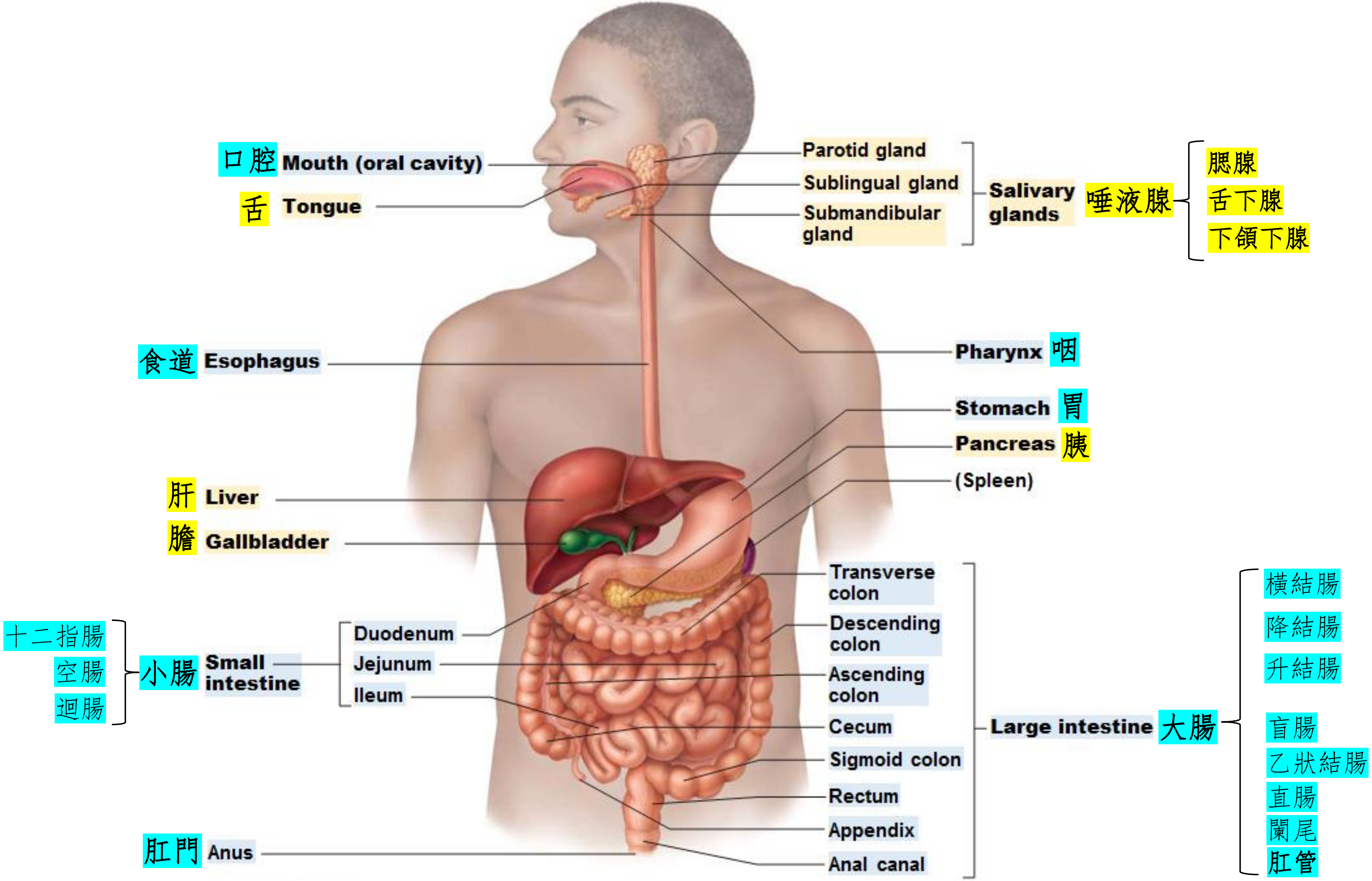


Overview of Digestive System
消化系統概述

- (1) **Alimentary canal 消化道**
- Continuous muscular tube 具有肌肉的連續管道
- From the mouth to anus 從口到肛門
- **Peristalsis 腸蠕動**
- **Site of digesting food消化食物的位置**:
  breaks down food into smaller molecules分解食物成更小的分子
- **Absorbs food molecules** into blood/lymph **將食物分子吸收**到血液/淋巴液中
- Organs器官:
  - **mouth口腔, pharynx咽部, esophagus食道,**
  - **Stomach胃,**
  - **small intestine小腸,**
  - **large intestine大腸, anus肛門**

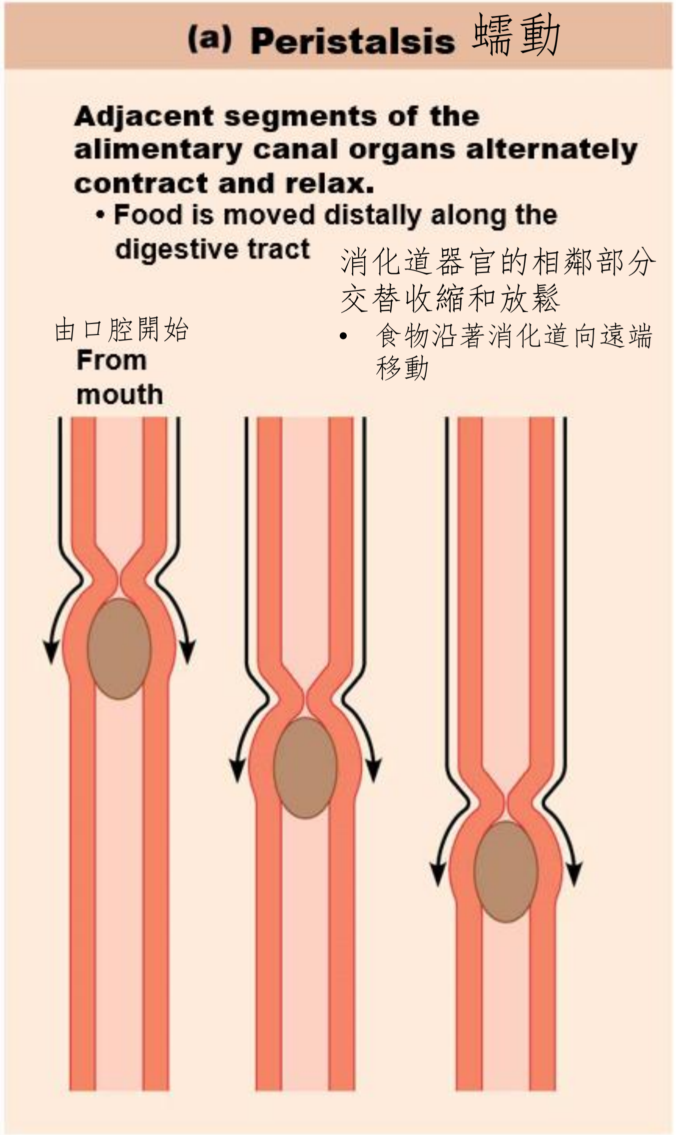
Overview of Digestive System
消化系統概述

- (2) **Accessory digestive organs輔助消化器官**
- **Teeth 牙齒**: mechanical breakdown of food 食物的機械性分解
- **Tongue 舌頭**: swallowing and sensations 吞咽和感覺
- **Gallbladder 膽囊**: storage and release of bile 膽汁的儲存和釋放
- **Digestive glands 消化腺**:
  produce secretions that help break down food chemically
  產生有助於化學性分解食物的分泌物
  - **Salivary glands 唾液腺**: produce **saliva**產生**唾液**
  - **Liver 肝**: produce **bile** 產生**膽汁**
  - **Pancreas 胰腺**: produce **pancreatic juice** 產生**胰液**


口腔 Mouth (oral cavity)
舌 Tongue
Parotid gland
Sublingual gland
Submandibular gland
Salivary glands 唾液腺
腮腺
舌下腺
下頜下腺
食道 Esophagus
Pharynx 咽
Stomach 胃
Pancreas 胰
(Spleen)
肝 Liver
膽 Gallbladder
十二指腸
空腸
迴腸
小腸 Small intestine
Duodenum
Jejunum
Ileum
Transverse colon
Descending colon
Ascending colon
Cecum
Sigmoid colon
Rectum
Appendix
Anal canal
Large intestine 大腸
橫結腸
降結腸
升結腸
盲腸
乙狀結腸
直腸
闌尾
肛管
肛門 Anus

Overview of Digestive System
消化系統概述
- Digestive Processes消化過程

- The processing of food by the digestive system involves 6 activities: 消化系統對食物的處理過程涉及6項活動：
- (1) **Ingestion攝入**:
  eating, taking food via the mouth通過嘴進食
- (2) **Propulsion推進**:
  movement of food from mouth to anus
  食物從口腔到肛門的移動
  - **Swallowing吞咽**
  - **Peristalsis蠕動**

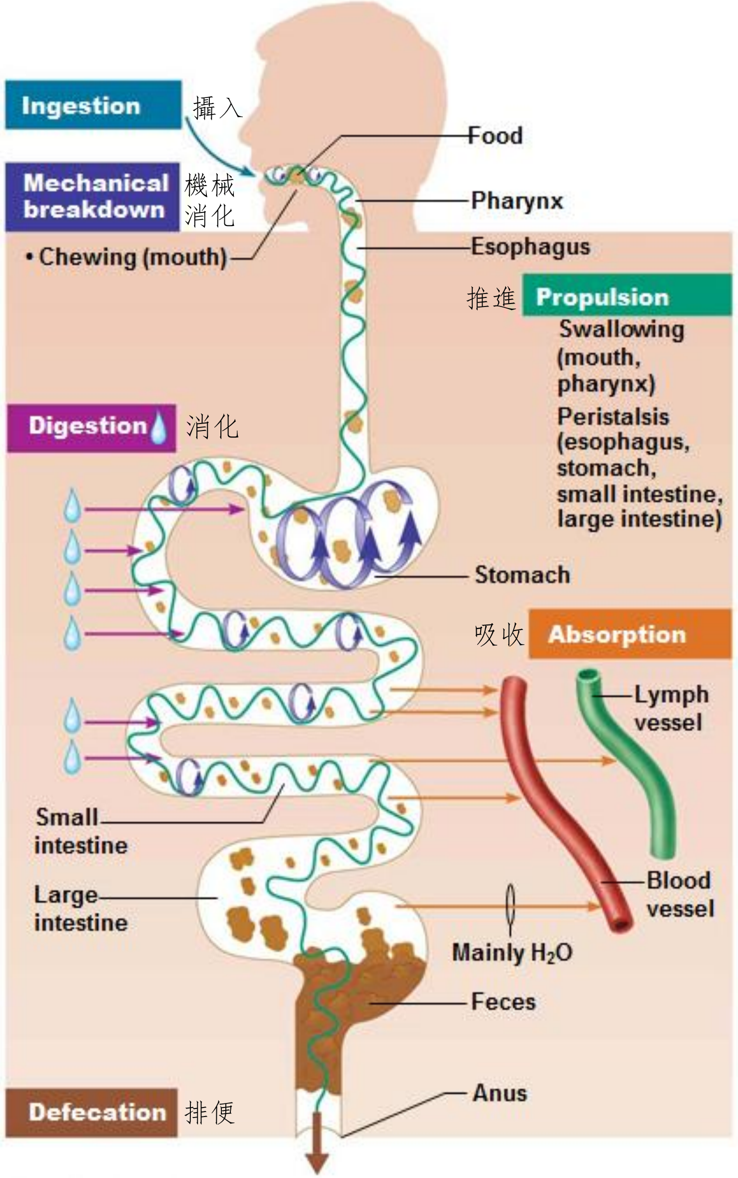


Overview of Digestive System消化系統概述
Digestive Processes消化過程

- The processing of food by the digestive system involves 6 activities:
消化系統對食物的處理過程涉及6項活動:
- (3) **Mechanical digestion 機械性消化**:
physically breaking food into small pieces for chemical digestion
將食物物理分解成小塊以進行化學消化
    - **Chewing咀嚼**
    - **Mixing food with digestive juice將食物與消化液混合**
- (4) **Chemical digestion 化學性消化**:
catabolic reactions enhanced by enzymes 透過酶增強的分解代謝反應
    - break down complex food molecules into simple chemical building blocks 將複雜的食物分子分解成簡單的化學構件

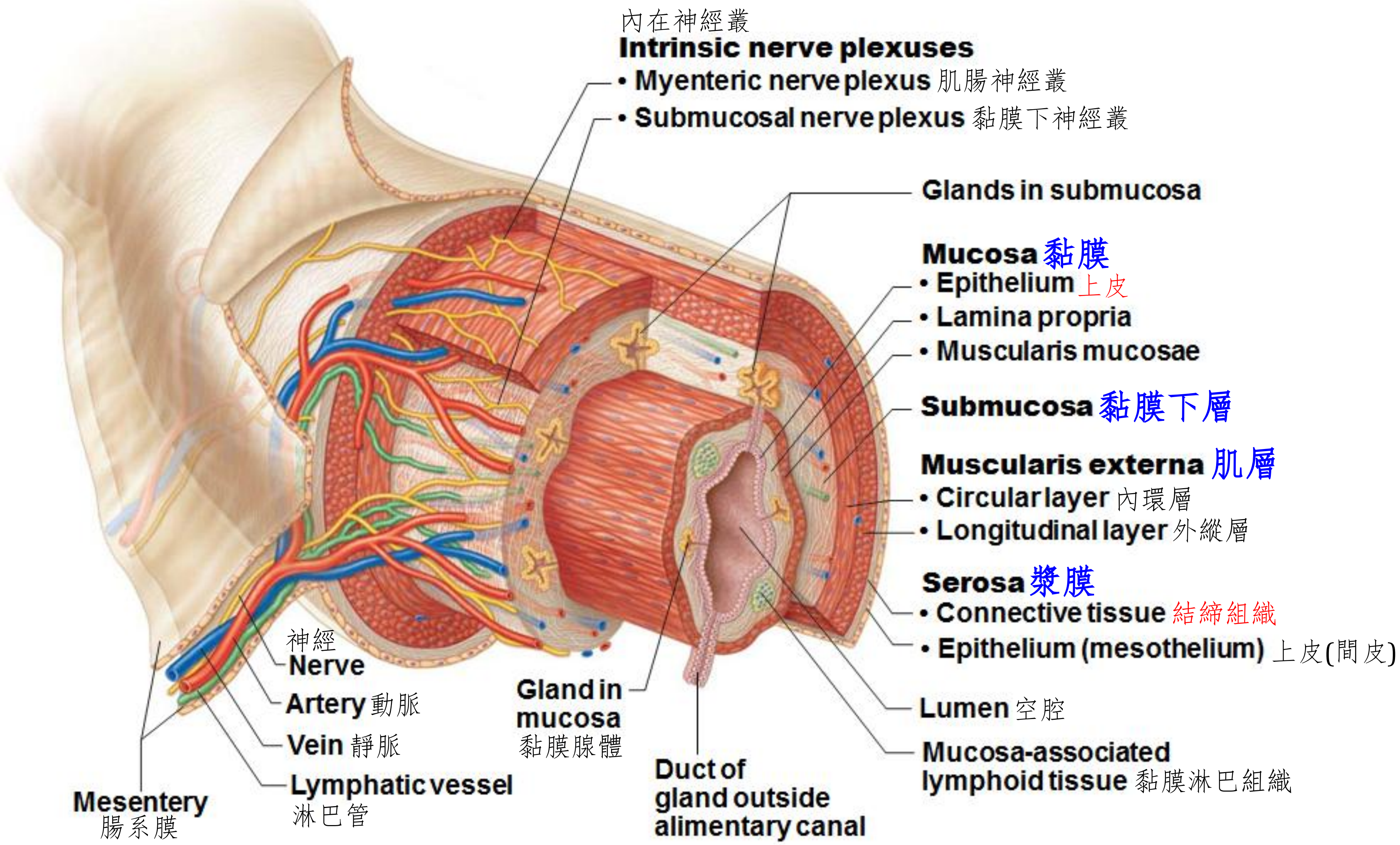
Overview of Digestive System
消化系統概述
Digestive Processes消化過程

- The processing of food by the digestive system involves 6 activities:消化系統對食物的處理過程涉及6項活動:
- (5) **Absorption 吸收**:
  passage of digested molecules from lumen of GI tract into blood or lymph
  經消化的分子從胃腸道空腔進入血液或淋巴液
- (6) **Defecation 排便**:
  elimination of indigestible substances via anus in form of **feces**
  通過肛門以**糞便**形式排除不能消化的物質



Overview of Digestive System消化系統概述
Basic Structure of Alimentary Canal 消化道的基本結構

- Walls of alimentary canal have the same four basic tissue layers 消化道壁具有相同的四個基本組織層
- **Mucosa 黏膜**
  - Epithelium that covers surface of lumen 覆蓋管腔表面的上皮
  - Secretes mucus, digestive enzymes, and hormones 分泌粘液、消化酶和激素
  - Absorbs end products of digestion吸收消化的最終產物
  - Protects against infectious disease 預防傳染病
- **Submucosa 黏膜下層**
  - Connective tissue 結締組織
  - Contains blood and lymphatic vessels, and nerve plexus含有血管、淋巴管以及神經叢
- **Muscularis externa 肌層**
  - Inner circular layer and outer longitudinal layer of smooth muscle tissue for peristalsis 用於蠕動的內環層和外縱層平滑肌組織
- **Serosa 漿膜**
  - Outermost connective tissue 最外層結締組織



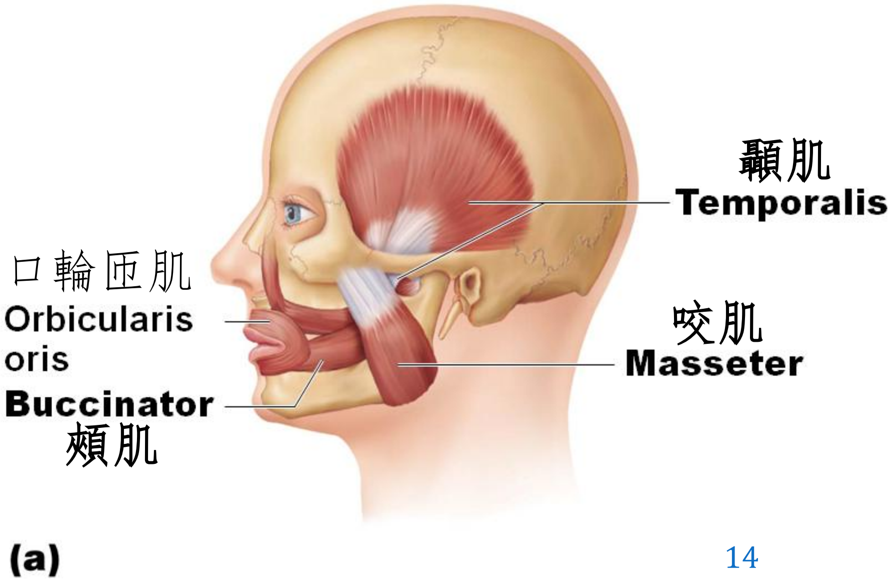
內在神經叢
Intrinsic nerve plexuses
• Myenteric nerve plexus 肌腸神經叢
• Submucosal nerve plexus 黏膜下神經叢
Glands in submucosa
Mucosa 黏膜
• Epithelium 上皮
• Lamina propria
• Muscularis mucosae
Submucosa 黏膜下層
Muscularis externa 肌層
• Circular layer 內環層
• Longitudinal layer 外縱層
Serosa 漿膜
• Connective tissue 結締組織
• Epithelium (mesothelium) 上皮(間皮)
Lumen 空腔
Mucosa-associated lymphoid tissue 黏膜淋巴組織
Duct of gland outside alimentary canal
Gland in mucosa 黏膜腺體
神經 Nerve
Artery 動脈
Vein 靜脈
Lymphatic vessel 淋巴管
Mesentery 腸系膜

Anatomy of Digestive System
消化系統解剖

- Mouth and Associated Organs口腔和相關器官
- **Mouth** (Oral cavity / Buccal cavity) is where food is **ingested**, **chewed**, and mixed with saliva that begins process of **digestion**, and **swallowing** process is initiated
  口腔(頰腔)是**攝入**、**咀嚼**、食物與唾液混合的地方，
  由此開始**消化**過程，並觸發**吞咽**過程
- Associated organs include:相關器官包括：
  - Tongue 舌
  - Salivary glands唾液腺
  - Teeth牙

Name of Muscle 肌肉名稱	Description描述	Origin (O)起點	Insertion (I)止點	Action活動
Masseter 咬肌	Powerful muscle that covers lateral aspect of mandibular ramus 覆蓋下頜支外側的強力肌肉	Zygomatic arch and zygomatic bone 顴弓和顴骨	Angle and ramus of mandible 下頜角和下頜支	Prime mover of jaw closure; elevates mandible 閉合下巴的主要肌肉；將下頜骨上提
Temporalis 顳肌	Fan-shaped muscle that covers parts of the temporal, frontal, and parietal bones 覆蓋部分顳骨、額骨和頂骨的扇形肌肉	Temporal fossa 顳窩	Coronoid process of mandible via a tendon that passes deep to zygomatic arch 通過顴弓下穿過的肌腱連接的下頜骨冠狀突	Closes jaw; elevates and retracts mandible; maintains the position of mandible at rest; deep anterior part may help protract mandible 閉合下巴；抬高和後縮下頜骨；在靜止時保持下頜骨的位置；深前部可能有助於前伸下頜骨
Buccinator 頰肌	Thin, horizontal cheek muscle; principal muscle of cheek; deep to masseter 薄而水平橫向的臉頰肌肉；臉頰的主要肌肉；在咬肌的深部	Molar region of maxilla and mandible 上頜骨和下頜骨的牙槽突	Orbicularis oris 口輪匝肌	Compresses the cheek; keeps food between grinding surfaces of teeth during chewing 擠壓臉頰；咀嚼時將食物放在牙齒的研磨面之間

Muscles of Mastication
咀嚼肌群



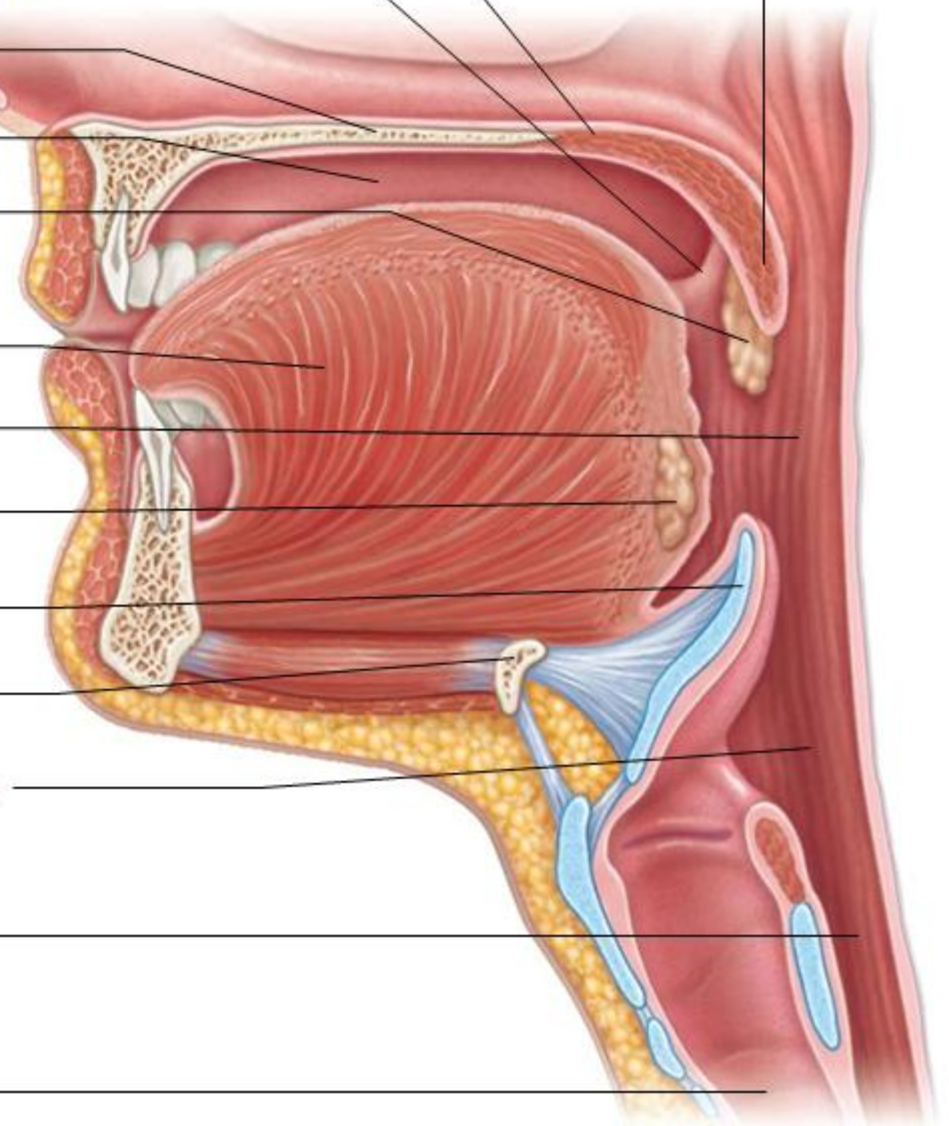
(Self-Study自學)

Closes off nasopharynx during swallowing
吞咽時閉合鼻咽部

Teeth:
Chewing to tear and grind food into smaller fragments
20 milk teeth
32 permanent teeth
牙齒：
咀嚼將食物撕碎並研磨成更小的碎片
20顆乳牙
32顆恆牙

Gripping, positioning, and mixing of food during chewing
Mixing food and saliva to form bolus
Involving in swallowing, speech, and taste
咀嚼過程中抓取、定位和混合食物
將食物和唾液混合形成食團
涉及吞咽、言語和味覺

Closes off trachea during swallowing
吞咽時閉合氣管

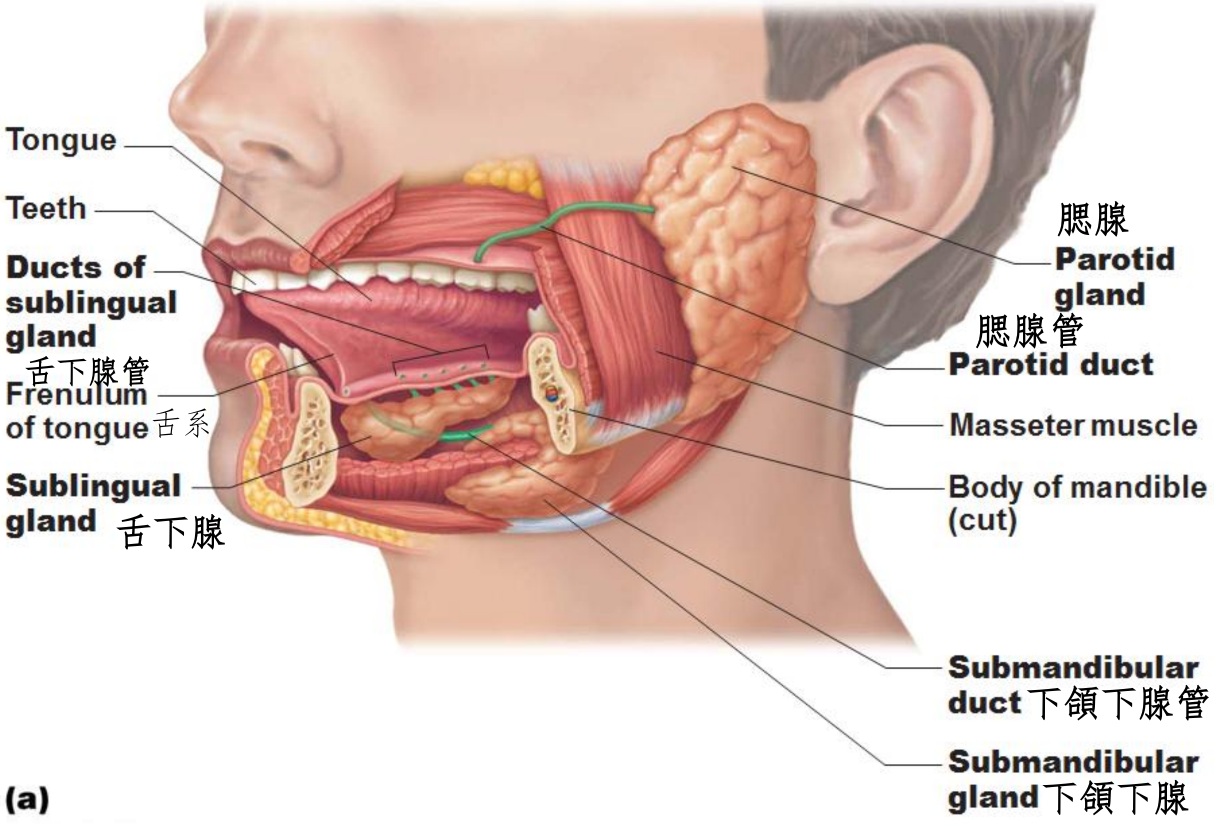


Functions of **saliva唾液**的作用

- **Cleanses mouth 清潔口腔**
- **Dissolves food chemicals** for taste
  溶解食物的化學物質以感受味覺
- **Moistens food**; compacts into bolus
  潤濕食物並將其壓縮成食團
- Begins breakdown of starch with enzyme **salivary amylase**
  用**唾液澱粉酶**開始分解澱粉

Anything that inhibits saliva secretion can promote tooth decay
任何抑制唾液分泌的事項都會促進蛀牙

Lack of saliva also makes it difficult to talk and swallow
缺乏唾液也會使說話和吞咽困難

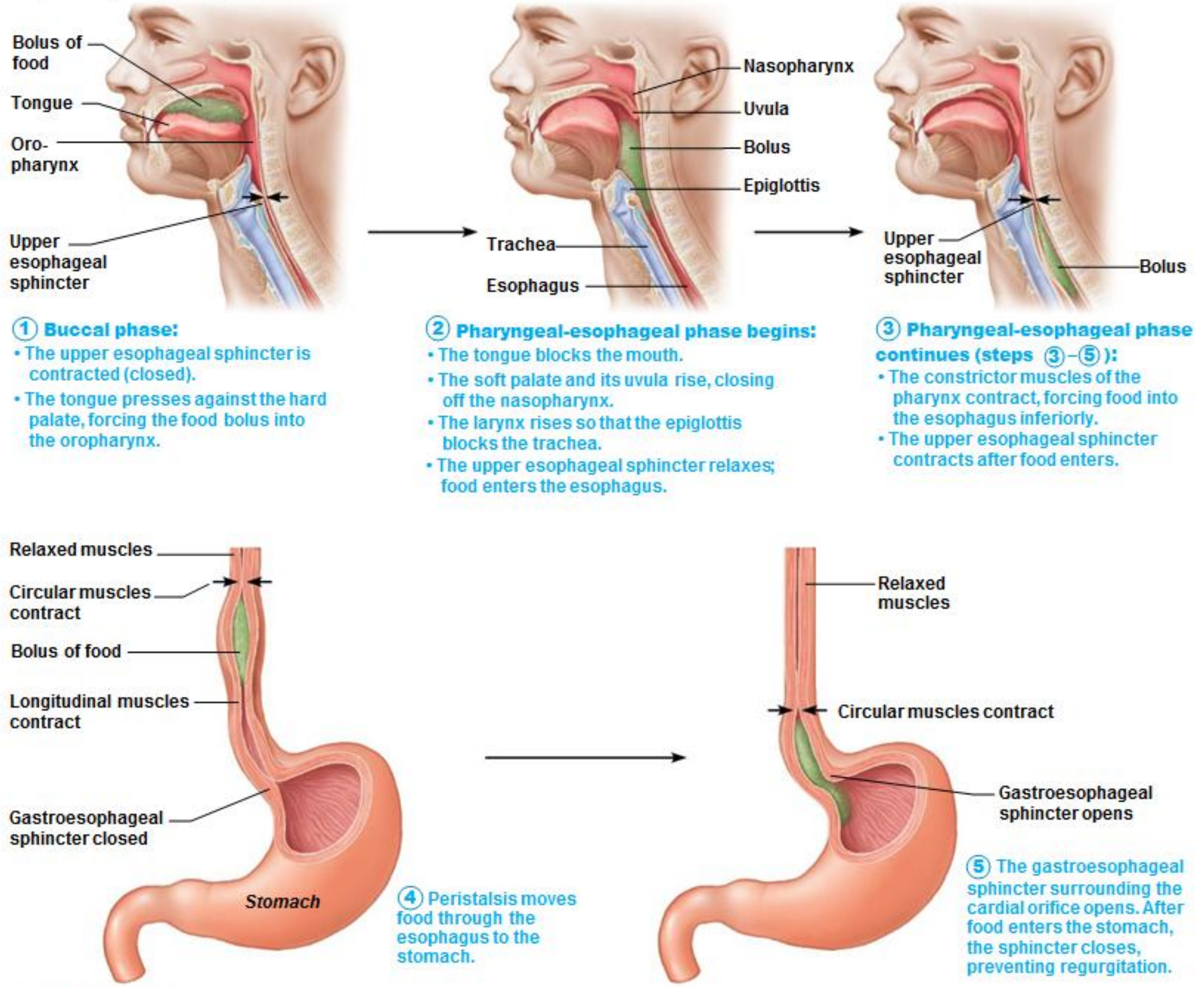


Anatomy of Digestive System
消化系統解剖

- Pharynx 咽
  - Connects to **esophagus** and **trachea** 連接**食道**和**氣管**
  - Allows passage of **food, fluids**, and **air** 允許**食物、液體**和**空氣**通過
- Esophagus 食道
  - Muscular tube that runs **from pharynx to stomach 從咽部延伸到胃**的肌肉管
  - **Peristalsis** for squeezing bolus toward stomach **蠕動**將食團擠向胃部
  - Secrete **mucus** to enhance bolus movement 分泌**粘液**以增強食團移動
  - **Heartburn 燒心**:
    Discomfort burning sensation caused by stomach acid regurgitating into esophagus
    胃酸反流到食道引起的不適燒灼感

Figure 23.14 Deglutition (swallowing).

(Self-Study 自學)



①頰期：
- 食管上括約肌收縮(閉合)。
- 舌頭上壓硬齶，迫使食團進入口咽部。

②咽食管期開始：
- 舌頭阻擋嘴巴。
- 軟齶及其懸雍垂上升以關閉鼻咽部。
- 喉部上升，使會厭阻塞氣管。
- 食管上括約肌放鬆；食團進入食道。

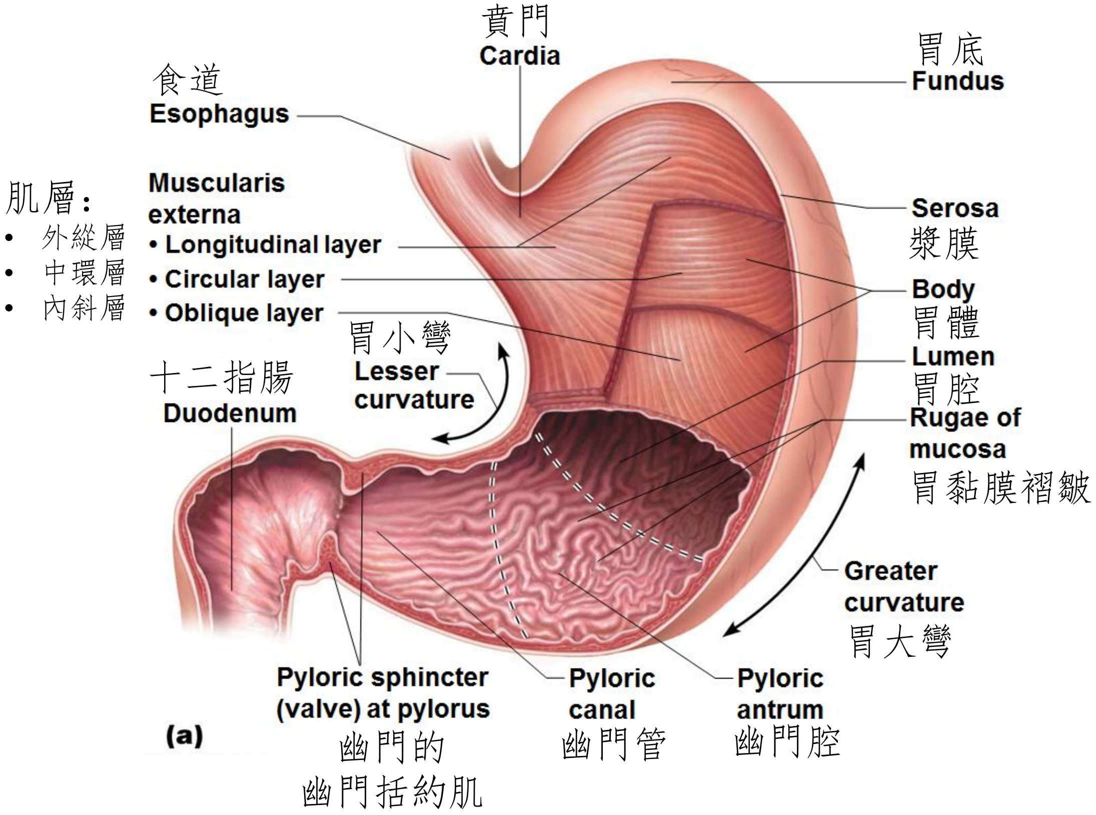
③咽食管期 (步驟③-⑤)：
- 咽部的收縮肌收縮，迫使食物向下進入食道。
- 食團進入後食管上括約肌收縮。

④蠕動將食團由食道移動到胃

⑤圍繞賁門的胃食管括約肌打開，食團進入胃後，括約肌閉合以防止反流。

Anatomy of Digestive System
消化系統解剖

- Stomach胃
- **temporary storage tank**:**臨時儲藏**:
  empty-50 ml; expand-4 L
  空-50毫升;膨脹-4公升
- starts chemical breakdown of **protein digestion**
  開始**蛋白質消化**的化學分解
- converts bolus to **chyme**
  將食團轉化為**食糜**

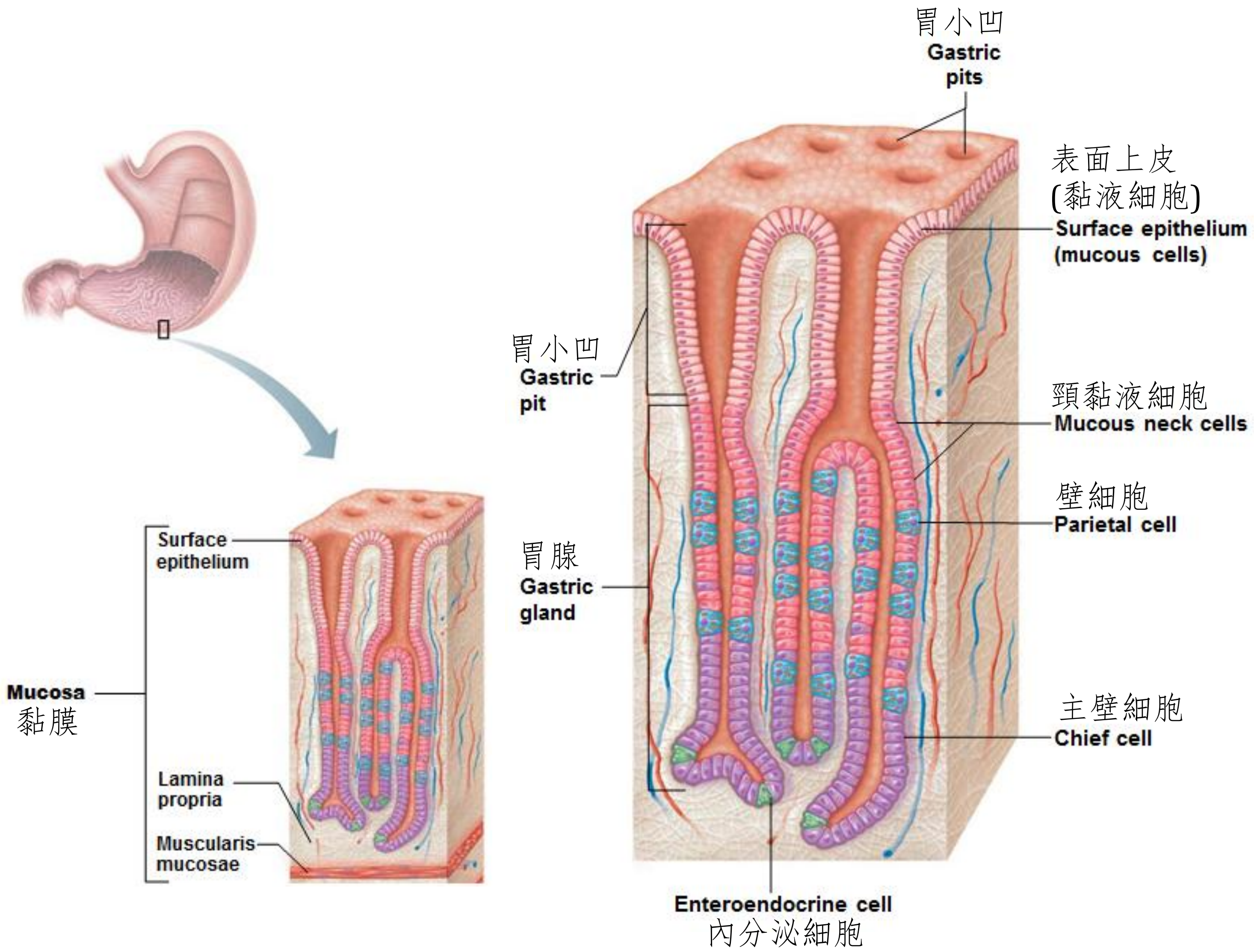


Secretions of gastric gland 胃腺分泌物：

- **Mucus粘液**: Protection 保護

- **Gastric juice胃液**:
  - **Pepsin胃蛋白酶**:
    Protein digestion蛋白質消化
  - **Hydrochloric acid (HCl)鹽酸**:
    Activate pepsin激活胃蛋白酶;
    Kill microorganisms in food
    殺死食物中的微生物

- **Intrinsic factor內在因子**:
  absorption of B12 in small intestine
  小腸對B12的吸收



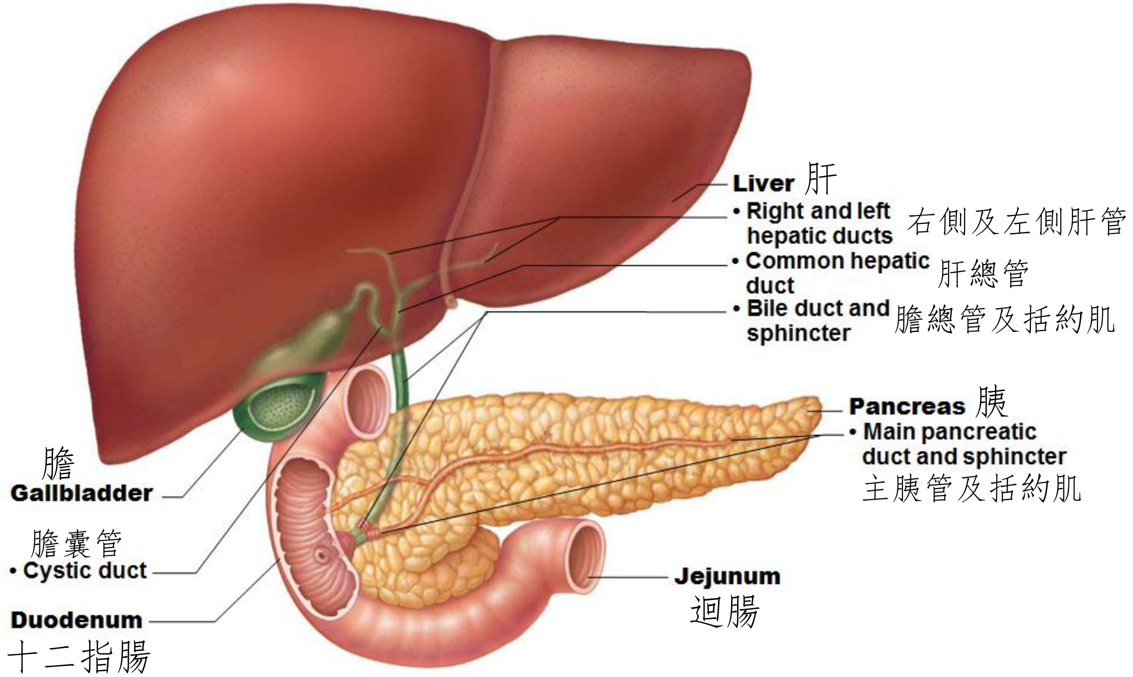
Anatomy of Digestive System
消化系統解剖

- Summary: functions of stomach 摘要： 胃的功能
  - Serves as **holding area for food** 作為**食物的存放區**
  - Churn bolus to **chyme** 攪動食團成**食糜**
  - Delivers small portion of chyme to small intestine將少量食糜輸送到小腸
  - **Kill microbes** by HCl 用鹽酸殺死微生物
  - Pepsin carries out **enzymatic digestion of proteins** 胃蛋白酶對**蛋白質**進行**消化**
  - Lipid-soluble alcohol and aspirin are absorbed into blood
    脂溶性酒精和阿司匹林被吸收到血液中
  - Secretes **intrinsic factor** for vitamin B12 absorption分泌內在因子以吸收維生素B12
  - Secretes **hormone (gastrin)** into blood 將激素**(胃泌素)**分泌到血液中

Anatomy of Digestive System
消化系統解剖

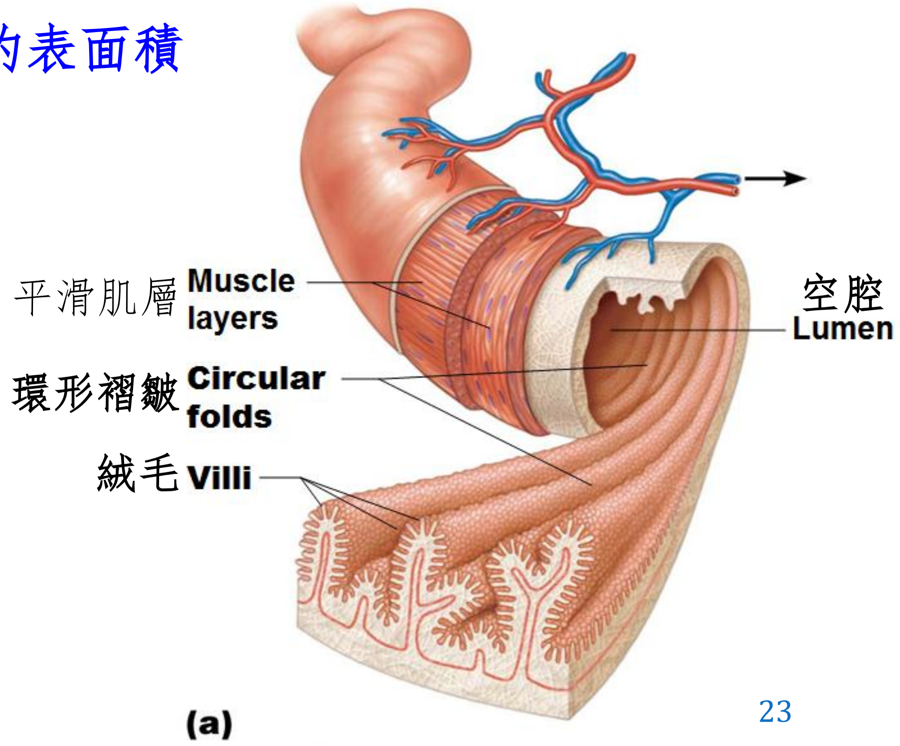
- Small intestine 小腸
  - major organ of digestion and absorption 消化吸收的主要器官
  - Length of 2–4 m; Diameter of 2.5–4 cm 長 2-4 m;直徑 2.5–4 厘米
  - 3 sections 3 部分：
    - **Duodenum十二指腸：**

      ~25.0 cm long 約25厘米長
    - **Jejunum空腸**
    - **Ileum迴腸**



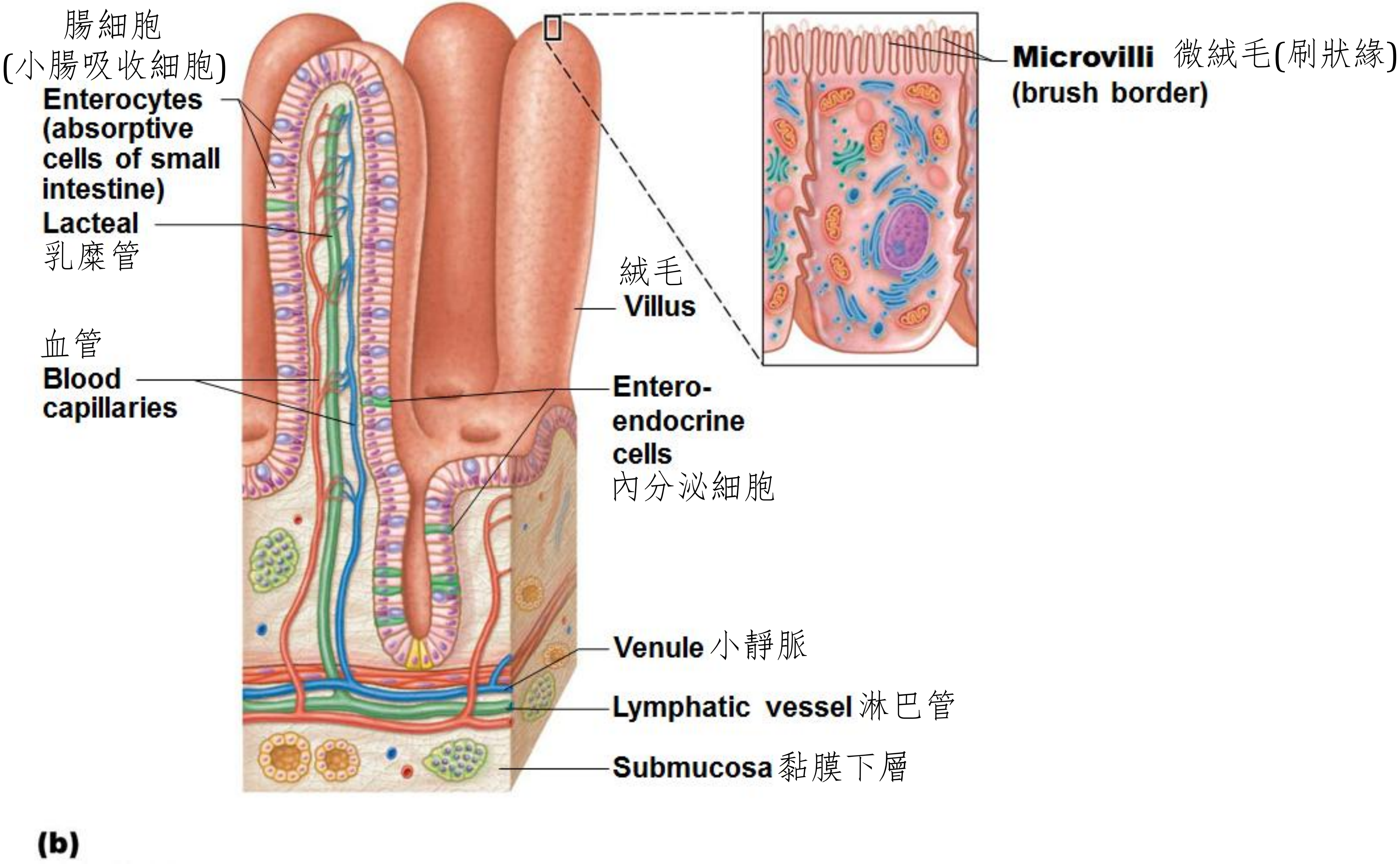
Anatomy of Digestive System
消化系統解剖

- Small intestine小腸
  - Small intestine's length and other structural modifications provide **huge surface area** for nutrient **absorption**
    小腸的長度和其結構變化為**吸收**營養提供了**巨大的表面積**
  - Modifications include變化包括：
    - Circular folds環形褶皺
    - **Villi 絨毛**
    - **Microvilli 微絨毛**



Anatomy of Digestive System
消化系統解剖

- Circular folds 環形褶皺
  - ~1 cm deep folds that force chyme to pass slowly through lumen, allowing more time for absorption 約1厘米深的褶皺迫使食糜緩慢穿過管腔，從而有更多時間進行吸收
- **Villi 絨毛**
  - ~1 mm high fingerlike projections of mucosa 黏膜上約1毫米高的手指狀突起
  - contains **dense capillary bed** and lymphatic capillary called **lacteal** for **absorption** 含有**緻密的毛細血管床**和稱為**乳糜管**的淋巴毛細血管，用於**吸收**
- **Microvilli 微絨毛**
  - Extensions of mucosal cell's plasma membrane that give fuzzy microscopic appearance called the brush border
    黏膜細胞的細胞膜之延伸部分，呈現模糊的微觀外觀，稱為刷狀緣
  - contains **brush border enzymes** for **final digestion of carbohydrate and protein** 含有**刷狀緣酶**，用於**碳水化合物和蛋白質的最終消化**

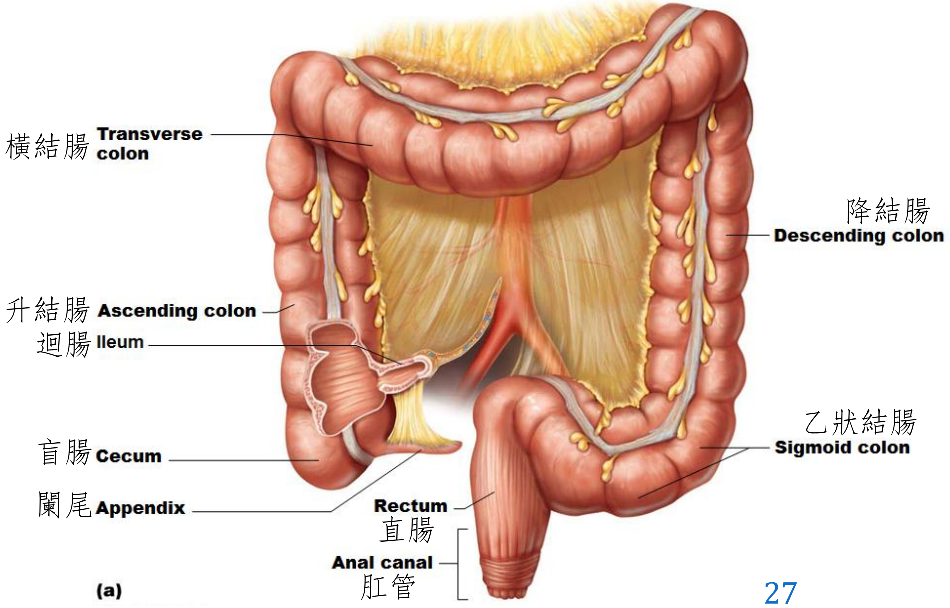

腸細胞
(小腸吸收細胞)
Enterocytes
(absorptive
cells of small
intestine)
Lacteal
乳糜管
血管
Blood
capillaries
絨毛
Villus
Entero-
endocrine
cells
內分泌細胞
Venule 小靜脈
Lymphatic vessel 淋巴管
Submucosa 黏膜下層
Microvilli 微絨毛(刷狀緣)
(brush border)
(b)

Anatomy of Digestive System
消化系統解剖

- Large intestine大腸
- Segments of large intestine大腸的不同分段
  - (1) **Cecum 盲腸**: first part of large intestine 大腸第一部分
  - (2) **Appendix 闌尾**: masses of lymphoid tissue 淋巴組織塊
  - (3) **Colon 結腸**:
    - ➢ **Ascending colon**: feces travels up right side of abdominal cavity
      升結腸：在腹腔右側向上運送糞便
    - ➢ **Transverse colon**: feces travels across abdominal cavity to left side
      橫結腸：糞便橫向穿過腹腔到左側
    - ➢ **Descending colon**: feces travels down left side of abdominal cavity
      降結腸：在腹腔左側向下運送糞便
    - ➢ **Sigmoid colon**: S-shaped portion that feces travels to pelvic cavity
      乙狀結腸：糞便進入盆腔的乙形部分

Anatomy of Digestive System
消化系統解剖

- Large intestine大腸
- Segments of large intestine大腸的不同分段
  - (4) **Rectum 直腸**: temporary storage of feces 暫存糞便
  - (5) **Anal canal 肛管**: last segment of large intestine that opens to body exterior at anus 大腸的最後一段，在肛門處向身體外部開放
    - Internal anal sphincter肛門內括約肌 (smooth muscle平滑肌)
    - External anal sphincter肛門外括約肌 (skeletal muscle骨骼肌)



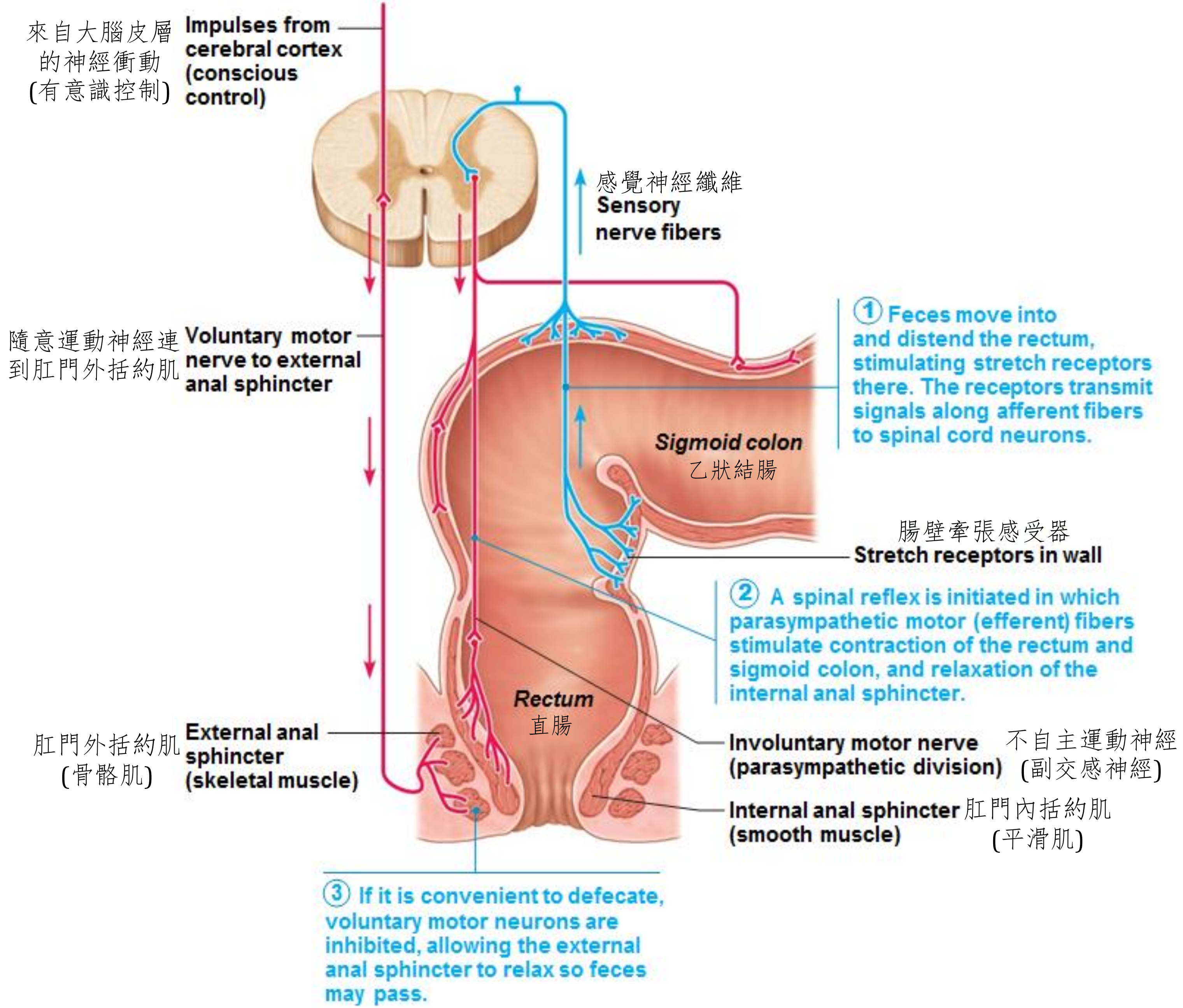
Anatomy of Digestive System
消化系統解剖

- Large intestine大腸
  - Does not contain circular folds, villi, or digestive secretions
    沒有環形褶皺、絨毛或消化分泌物
  - Contains many mucus-producing goblet cells含有許多產生粘液的杯狀細胞
  - Residue remains in large intestine 12–24 hours食物殘渣在大腸中殘留 12-24 小時
  - No food breakdown occurs except what enteric bacteria digest
    除了腸道細菌消化的食物外，不會發生食物分解
  - Vitamins (made by bacterial flora), **water**, and **electrolytes** (especially Na^{+} and Cl^{-}) are **absorbed**
    維生素(由菌群產生)、**水和電解質**(尤其是 Na^{+} 和 Cl^{-})**被吸收**
  - Major functions of large intestine is **propulsion of feces to anus** and **defecation**
    大腸的主要功能是**將糞便推進到肛門**和**排便**

Anatomy of Digestive System
消化系統解剖

- **Defecation 排便**
  1. **Feces movements** toward rectum by peristalsis 蠕動使**糞便**向直腸**移動**
     Distension of rectal wall triggers **defecation reflex**
     直腸壁擴張引發**排便反射**
  2. Parasympathetic signals stimulate **contraction of sigmoid colon and rectum**
     副交感神經信號刺激**乙狀結腸和直腸的收縮**
     Relaxation of internal anal sphincter 肛門內括約肌放鬆
  3. Conscious control allows **relaxation of external anal sphincter**
     有意識的控制**放鬆肛門外括約肌**
     Muscles of rectum contract to **expel feces**
     直腸肌肉收縮**排出糞便**

(Self-Study自學)



①糞便進入直腸並擴張直腸，刺激那裡的牽張感受器，牽張感受器沿著傳入纖維將信號傳遞到脊髓神經元。

②脊髓反射啟動，副交感神經運動纖維刺激直腸和乙狀結腸的收縮，以及肛門內括約肌的放鬆。

③如果方便排便，隨意運動神經元就會受到抑制，使肛門外括約肌放鬆，從而使糞便可以通過。

Physiology of Digestion and Absorption 消化和吸收生理學

- **Chemical Digestion化學性消化**
  - Breaks down ingested foods into their chemical building blocks 將攝入的食物分解成其化學組成部分
  - **Digestive enzymes** are involved涉及**消化酶**
  - Final end products are small enough to be absorbed across wall of small intestine 最終產品足夠小到可以穿過小腸壁吸收

- **Absorption吸收**
  - Moves substances from lumen of gut into body 將物質從腸腔轉移到體內
  - Most nutrients are absorbed before chyme reaches ileum 大多數營養物質在食糜到達迴腸之前被吸收
  - **Passive and active ways of membrane transport** are use 使用**被動和主動膜運輸方式**

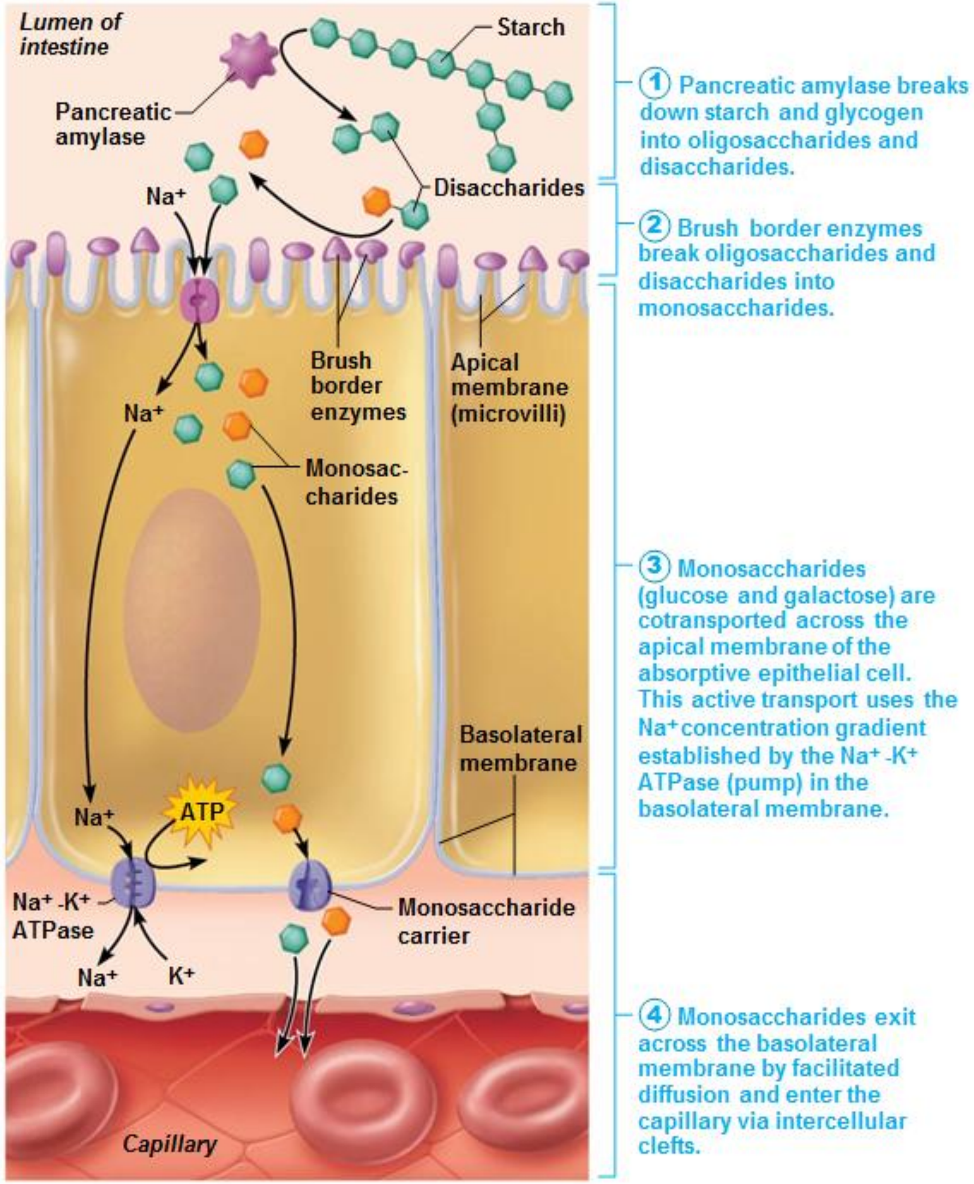
Physiology of Digestion and Absorption 消化和吸收生理學

- Digestive juices in small intestine 小腸內的消化液：
- **Bile膽汁**
  - Produce by liver; Store by gallbladder; Release into **duodenum**
    肝臟生產；膽囊儲存；被釋放到**十二指腸**
  - Contains **bile acid and phospholipids** for **emulsifying lipids**
    含有**膽酸和磷脂**，用於**乳化脂質**
  - Contains **bicarbonate** for **neutralizing gastric acid**
    含有用於**中和胃酸**的**碳酸氫鹽**
  - Other components: **bile pigment**, **cholesterols**, ions 其他成分：**膽汁色素**、**膽固醇**、離子
- **Pancreatic juice胰液**
  - Produce by pancreas; Release into **duodenum** 生產胰腺；被釋放到**十二指腸**
  - Contains **digestive enzymes** for **chemical digestion of carbohydrate, proteins, and lipids** 含有**消化酶**，用於**碳水化合物、蛋白質和脂質的化學消化**
  - Contains **bicarbonate** for **neutralizing gastric acid**
    含有用於**中和胃酸**的**碳酸氫鹽**

Physiology of Digestion and Absorption
消化和吸收生理學

- Digestion and Absorption of Carbohydrates碳水化合物的消化和吸收

Enzyme 酶	Site of action 作用位置	Source of enzyme 酶的來源	Substrate 底物	Products 產物
Salivary Amylase 唾液澱粉酶	Oral cavity 口腔	Saliva 唾液	5% Starch澱粉 / Glycogen糖原	Maltose 麥芽糖
Pancreatic Amylase 胰澱粉酶	Duodenum 十二指腸	Pancreatic juice 胰液	95% Starch澱粉 / Glycogen糖原	Maltose 麥芽糖
Maltase 麥芽糖酶	Small intestine 小腸	Attached to microvilli “Brush border enzyme” 附著在微絨毛上 “刷狀緣酶”	Maltose 麥芽糖	Glucose葡萄糖 + Glucose葡萄糖
Sucrase 蔗糖酶			Sucrose 蔗糖	Glucose葡萄糖 + Fructose果糖
Lactase 乳糖酶			Lactose 乳糖	Glucose葡萄糖 + Galactose半乳糖



Monosaccharides are absorbed into _blood_ via _facilitated diffusion_ and _secondary active transport_!

單糖通過促進擴散和次級主動運輸被吸收到血液中！

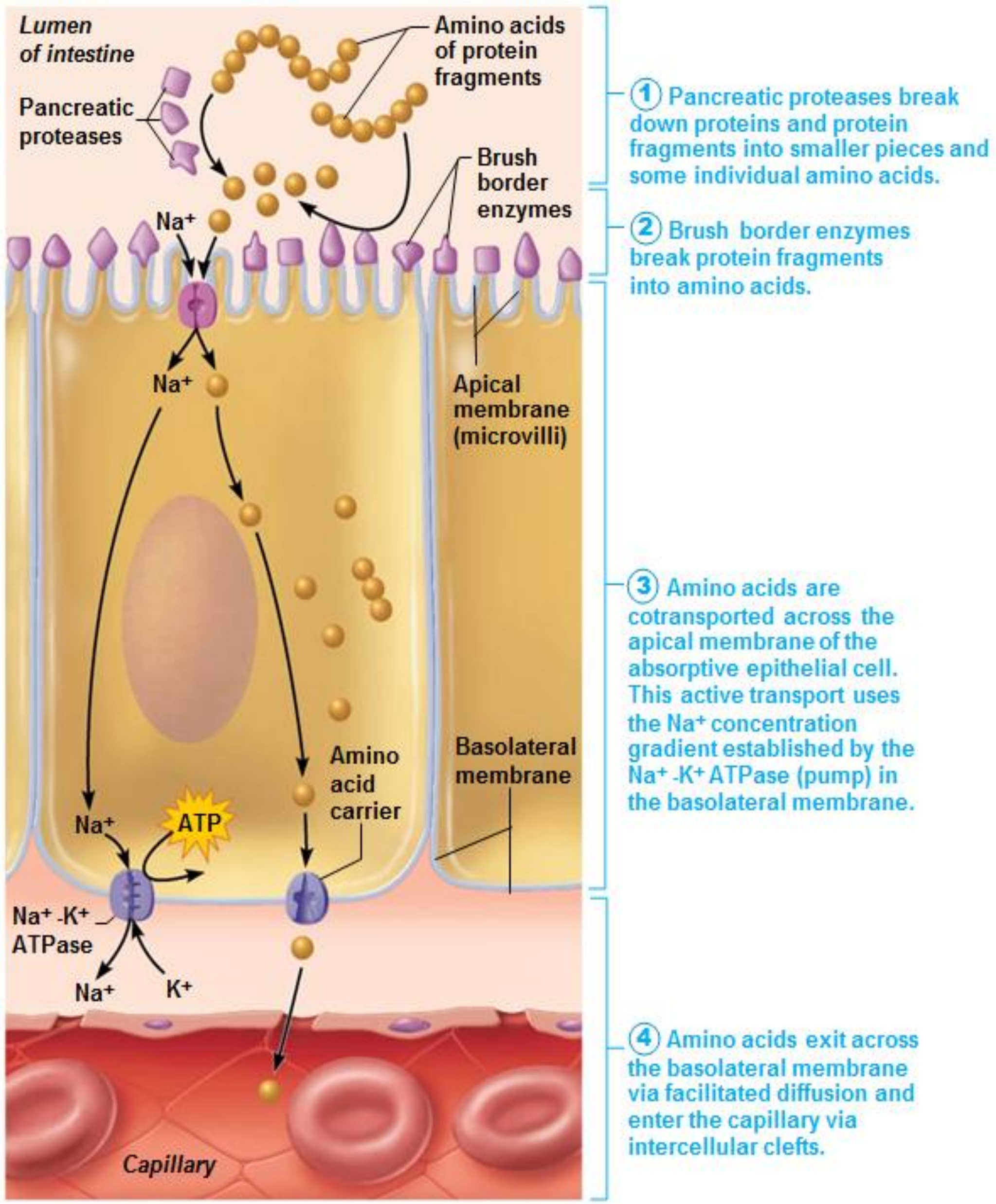
Physiology of Digestion and Absorption 消化和吸收生理學

- Digestion and Absorption of Proteins 蛋白質的消化和吸收

Enzyme 酶	Site of action 作用位置	Source of enzyme 酶的來源	Substrate 底物	Products 產物
Pepsin 胃蛋白酶	Stomach 胃	Gastric juice 胃液	Large polypeptides 大的多肽	Small polypeptides 小的多肽
Trypsin 胰蛋白酶, chymotrypsin 胰凝乳蛋白酶, carboxypeptidase 羧肽酶	Small intestine 小腸	Pancreatic juice 胰液	Small polypeptides 小的多肽	Amino acids 胺基酸
Aminopeptidase 氨肽酶	Small intestine 小腸	Brush border enzyme 刷狀緣酶	Small polypeptides 小的多肽	Amino acids 胺基酸

Amino acids are absorbed into *blood* via *facilitated diffusion* and *secondary active transport*!

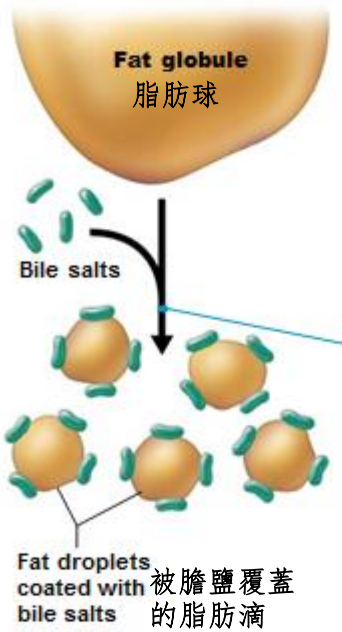
胺基酸通過促進擴散和次級主動運輸被吸收到血液中！



- In rare cases, intact proteins are taken up by intestinal epithelial cells by endocytosis and are released into body
  在極少數情況下，完整的蛋白質通過胞吞作用被腸上皮細胞吸收並釋放到體內
  - More common in patients with immature/weak intestinal mucosa such as newborn infants
    多見於腸黏膜未成熟/脆弱的患者，例如新生兒
  - May result in food allergies as immune system “sees” intact proteins as harmful and starts an attack
    可能導致食物過敏，因為免疫系統“認為”完整的蛋白質是有害的並開始攻擊
  - Allergies usually disappear as mucosa matures
    過敏通常會隨著黏膜的成熟而消失

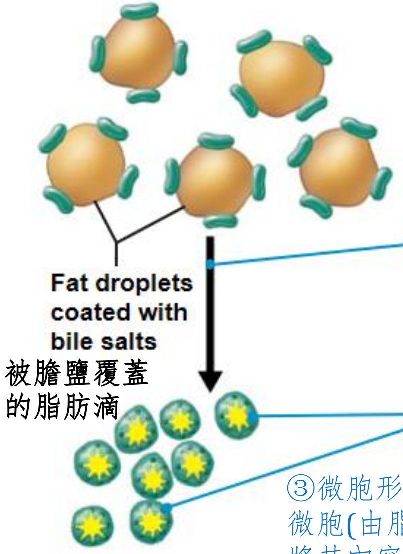
Physiology of Digestion and Absorption
消化和吸收生理學

- Digestion and Absorption of Lipids 脂肪的消化和吸收
- **Digestion of lipids** occurs mainly in **small intestine**.
  脂肪的消化主要發生在**小腸**中。
- **Bile** is essential for lipid emulsification.
  膽汁對於乳化脂肪至關重要。
- **Lipase in pancreatic juice** is essential for chemical digestion of triglyceride.
  胰液中的脂肪酶對於甘油三酯的化學消化至關重要。



① **Emulsification.** Bile salts in the duodenum break large fat globules into smaller fat droplets, increasing the surface area available to lipase enzymes.

①乳化：
十二指腸中的膽鹽將大脂肪球分解成較小的脂肪滴，從而增加脂肪酶可用的表面面積。

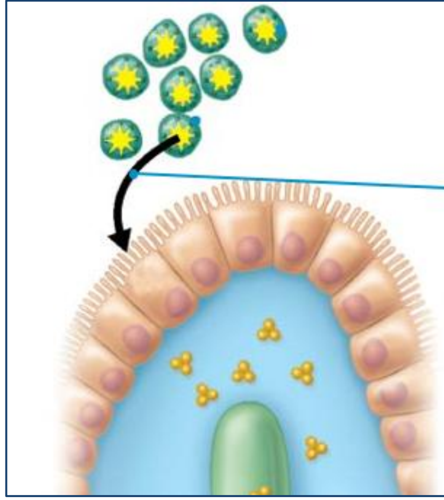


②消化：
胰脂肪酶水解甘油三酯，產生甘油單酯和遊離脂肪酸。

② **Digestion.** Pancreatic lipases hydrolyze triglycerides, yielding monoglycerides and free fatty acids.

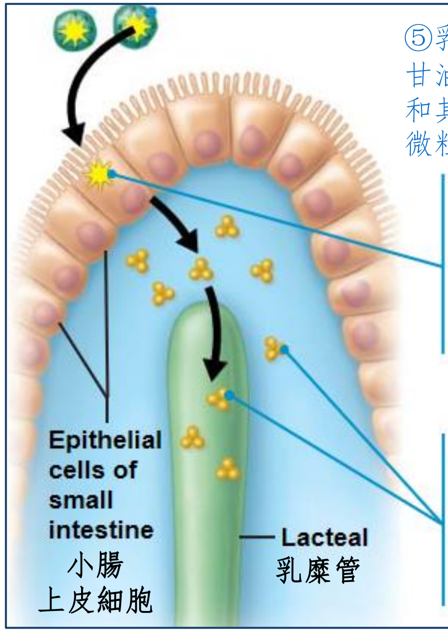
③ **Micelle formation.** Micelles (consisting of fatty acids, monoglycerides, and bile salts) ferry their contents to epithelial cells.

③微胞形成：
微胞(由脂肪酸、甘油單酯和膽鹽組成)將其內容物運送向上皮細胞。



④擴散：
甘油單酯和脂肪酸由微胞擴散到上皮細胞內。

④ **Diffusion.** Fatty acids and monoglycerides diffuse from micelles into epithelial cells.



⑤乳糜微粒形成：
甘油單酯和脂肪酸被重新組合，並和其脂肪及蛋白質一起包裝成乳糜微粒。

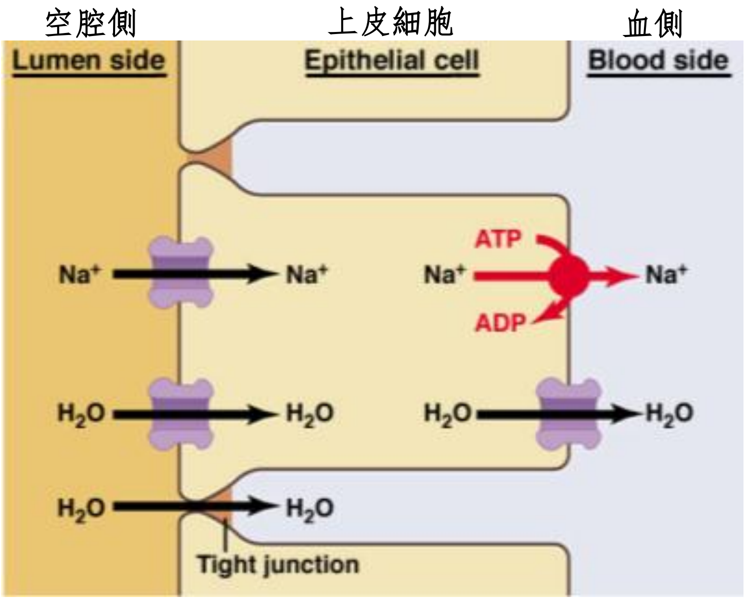
⑤ **Chylomicron formation.** Fatty acids and monoglycerides are recombined and packaged with other fatty substances and proteins to form chylomicrons.

⑥ **Chylomicron transport.** Chylomicrons are extruded from the epithelial cells by exocytosis, enter lacteals, and are carried away from the intestine in lymph.

⑥乳糜微粒運送：
乳糜微粒透過胞吐作用離開上皮細胞，進入乳糜管，並經淋巴液運送離開腸臟。

Physiology of Digestion and Absorption 消化和吸收生理學

- Absorption of water 水的吸收
  - 9 L water enter small intestine everyday 每天9L水進入小腸
  - Over 80% is absorbed in the **small intestine** by **osmosis** 超過80%通過**滲透**在**小腸**中吸收
  - Most of rest is absorbed in large intestine 其餘大部分被大腸吸收
  - Net osmosis occurs when difference of water potentials is made by active transport of solutes (Mainly Na^+) 當溶質(主要是 Na^+)的主動傳輸產生水勢的差異，就會發生淨滲透
  - →Water uptake is coupled with solute uptake →水分吸收與溶質吸收相結合



Physiology of Digestion and Absorption 消化和吸收生理學

- Absorption of Vitamins 維生素的吸收
- In small intestine在小腸
  - **Fat-soluble vitamins** (A, D, E, and K) are carried by micelles and absorbed by **simple diffusion**
    脂溶性維生素(A、D、E 和 K)由微胞攜帶並通過**簡單擴散**吸收
  - **Water-soluble vitamins** (C and B) are absorbed by **facilitated diffusion** or active transport
    水溶性維生素(C 和 B)通過**促進擴散**或主動轉運被吸收
  - **Vitamin B12** (large molecule) binds with intrinsic factor and is absorbed by **endocytosis**
    維生素B12(大分子)與內在因子結合，被**胞吞作用**吸收
- In large intestine在大腸
  - vitamin K and B vitamins made by friendly bacteria are absorbed
    由益菌產生的維生素K和B族維生素被吸收

(Self-Study自學)

Physiology of Digestion and Absorption 消化和吸收生理學

- Absorption of Minerals礦物質的吸收
- **Iron and calcium** are absorbed in **duodenum**
  鐵和鈣在**十二指腸**中被吸收
- Iron and calcium absorption is related to need
  鐵和鈣的吸收與需求有關
  - Iron ions are stored in mucosal cells. When needed, the ions are transported in blood
    鐵離子儲存在粘膜細胞中。需要時，離子在血液中運輸
  - Ca absorption is regulated by vitamin D and parathyroid hormone (PTH)
    鈣的吸收受維生素 D 和甲狀旁腺激素(PTH)的調節

Physiology of Digestion and Absorption 消化和吸收生理學

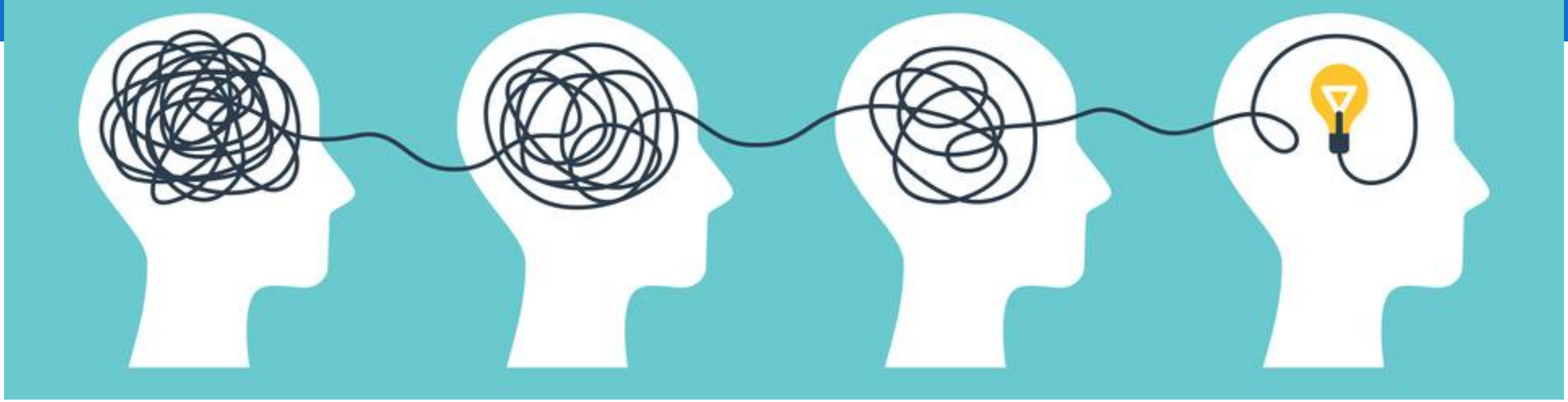
- Regulation of Digestive Secretion消化分泌的調節
- Secretion of digestive juices is controlled by both nervous activities and hormones. 消化液的分泌由神經活動和激素控制。
- **Neural regulation神經調節**
  - Generally stimulated by parasympathetic division and inhibited by sympathetic division of **autonomic nervous system (ANS)**
    通常受到**自主神經系統(ANS)**副交感神經的刺激和交感神經的抑制
- **Endocrine (hormonal) regulation 內分泌(激素)調節**
  - **Gastrin 胃泌素**
  - **Secretin 胰泌素**
  - **Cholecystokinin (CCK) 膽囊收縮素**
  - **Gastric inhibitory peptide (GIP) 胃抑肽**

(Self-Study自學)

Hormone 激素	Secreted by 分泌器官	Effects 效果
Gastrin 胃泌素	Stomach 胃	1. Stimulate the secretion of HCl 刺激鹽酸的分泌 2. Stimulate the secretion of pepsin 刺激胃蛋白酶的分泌
Secretin 胰泌素	Small intestine 小腸	1. Stimulate bicarbonate secretion in pancreatic juice and bile 刺激胰液和膽汁中的碳酸氫鹽分泌
Cholecystokinin (CCK) 膽囊收縮素	Small intestine 小腸	1. Stimulate secretion of pancreatic enzymes 刺激胰酶分泌 2. Stimulate contraction of gallbladder 刺激膽囊收縮 3. Inhibit gastric motility and secretion 抑制胃蠕動和分泌
Gastric inhibitory peptide (GIP) 胃抑肽	Small intestine 小腸	1. Stimulate secretion of insulin 刺激胰島素分泌 2. Inhibit gastric motility and secretion 抑制胃蠕動和分泌

Physiology of Digestion and Absorption
消化和吸收生理學

- Aging and the digestive system 衰老與消化系統
- GI tract activity declines with age 胃腸道活動隨著年齡的增長而下降
  - produces less digestive juice; 產生較少的消化液;
  - absorption is less efficient; 吸收效率較低;
  - peristalsis slows, causing decline in frequency of bowel movements 蠕動減慢，導致排便頻率下降
- Taste/smell becomes less acute; periodontal disease often develops 味覺/嗅覺變得不那麼敏銳;牙周病經常發展
- Constipation, fecal incontinence, and cancer of GI tract are fairly common 便秘、大便失禁和胃腸道癌相當常見
- Prevention: regular dental and medical examinations 預防：定期進行牙科和體檢



L5 Revision Exercise L5 複習練習

Screen this QR code to complete all questions in this revision exercise.
掃瞄此二維碼以完成
此複習練習中的所有問題。

References

- 梅盛平, & 徐国昌. (2022). 正常人体结构学. 重庆大学电子音像出版社有限公司.
- 周华, & 杨向群 (2022). 人体解剖生理学(第8版). 人民卫生出版社.
- 管又飞, 罗自强, & 薛明明主编. 医学生理学(第5版). (2024). 北京大学医学出版社.
- 中华人民共和国国家卫生健康委员会. 常用临床医学名词. (2023). http://www.nhc.gov.cn/yzygj/s7659/202403/bdb6d9140b734323b49c9764deba8c45.shtml
- Marieb, E.N., Hoehn, K. (2018) Human Anatomy & Physiology (11th ed.). Pearson International ed.
- Martini, F. H. (2017). Fundamentals of Anatomy and Physiology (11th ed.). England: Pearson Benjamin Cummings.
- Olson, T.R., Pawlina Woichiech (2015) A.D.A.M. Student Atlas of Anatomy (2nd ed.) Cambridge University Press. https://doi.org/10.1017/CBO9780511800160
- Snell, R.S. (2007) Clinical Anatomy by Systems. Lippinocott Williams & Wilkins